

```
/* assessment_cover.sql */
```

```
SELECT  knowledge, skills, results
FROM    'Kagiso Matenchi'
WHERE   topic = 'SQL Date Functions'
AND     platform = 'Databricks';
```

SQL DATE FUNCTIONS EXERCISE

STUDENT ASSESSMENT · SQL ON DATABRICKS

TOPIC

Date Functions

extract · format · calculate

QUESTIONS

20 + 1 BONUS

beginner-intermediate

PLATFORM

Databricks

SQL warehouse

KAGISO MATENCHI

SQL PRACTITIONER · BRIGHTLEARN

▶ RUN ON DATABRICKS · 11 JUNE 2026

-- end of cover page --

QUESTION 1:

The screenshot shows the Databricks SQL Editor interface. The left sidebar contains navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover Beta, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, and Data Ingestion. The main editor area displays a SQL query labeled --Q1:orders. The query selects order_id, customer_id, order_date, and DAYNAME(order_date) AS day_name from the orders table. Below the query, a table shows the results of the query, with 5 rows and 6 columns: order_id, customer_id, order_date, and day_name. The status bar at the bottom indicates 5 rows, 2.25s runtime, and a link to see performance.

```
--Q1:orders
SELECT
  order_id,
  customer_id,
  order_date,
  DAYNAME(order_date) AS day_name
FROM orders;
```

order_id	customer_id	order_date	day_name
1	1001	2026-05-01	Fri
2	1002	2026-05-02	Sat
3	1003	2026-05-03	Sun
4	1004	2026-05-04	Mon
5	1005	2026-05-05	Tue

5 rows | 2.25s runtime | See performance (1) | Checked for improvements | Refreshed now

QUESTION 2:

The screenshot shows the Databricks SQL Editor interface. The left sidebar contains navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover Beta, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, and Data Ingestion. The main editor area displays a SQL query labeled --Q2:. The query selects customer_id, customer_name, signup_date, and MONTHNAME(signup_date) AS signup_month_name from the customer_signups table. Below the query, a table shows the results of the query, with 5 rows and 5 columns: customer_id, customer_name, signup_date, and signup_month_name. The status bar at the bottom indicates 5 rows, 1.05s runtime, and a link to see performance.

```
--Q2:
SELECT
  customer_id,
  customer_name,
  signup_date,
  MONTHNAME(signup_date) AS signup_month_name
FROM customer_signups;
```

customer_id	customer_name	signup_date	signup_month_name
1	John	2026-01-15	Jan
2	Mary	2026-02-20	Feb
3	Peter	2026-03-05	Mar
4	Sarah	2026-04-18	Apr
5	Thabo	2026-05-30	May

5 rows | 1.05s runtime | See performance (1) | Checked for improvements | Refreshed now

QUESTION 3:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON + X

Run selected (1000) Just now (1) workspace default New SQL editor: ON

```
--Q3:sales
SELECT
  sale_id,
  product_name,
  sale_date,
  MONTH(sale_date) AS sale_month
FROM sales;
```

Add parameter

	sale_id	product_name	sale_date	sale_month
1	S001	Laptop	2026-01-10	1
2	S002	Mouse	2026-02-15	2
3	S003	Keyboard	2026-03-20	3
4	S004	Monitor	2026-04-25	4
5	S005	Desk	2026-05-30	5

5 rows | 0.81s runtime See performance (1) Finding improvements... Refreshed now

QUESTION 4:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON + X

Run selected (1000) Just now (1) workspace default New SQL editor: ON

```
--Q4:transactions
SELECT
  transaction_id,
  customer_id,
  transaction_date,
  YEAR(transaction_date) AS transaction_year
FROM transactions;
```

Add parameter

	transaction_id	customer_id	transaction_date	transaction_year
1	T001	101	2024-12-15	2024
2	T002	102	2025-01-20	2025
3	T003	103	2025-06-10	2025
4	T004	104	2026-02-05	2026
5	T005	105	2026-05-25	2026

5 rows | 0.85s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 5:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q5: deliveries
SELECT
  delivery_id,
  customer_id,
  delivery_date,
  DAY(delivery_date) AS day_of_month
FROM deliveries;
```

Add parameter

	delivery_id	customer_id	delivery_date	day_of_month
1	D001	101	2026-05-01	1
2	D002	102	2026-05-08	8
3	D003	103	2026-05-15	15
4	D004	104	2026-05-22	22
5	D005	105	2026-05-29	29

5 rows | 0.80s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 6:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q6: employees
SELECT
  employee_id,
  employee_name,
  department,
  CURRENT_DATE() AS today_date
FROM employees;
```

Add parameter

	employee_id	employee_name	department	today_date
1	301	Nandi	Sales	2026-06-11
2	302	Brian	IT	2026-06-11
3	303	Lerato	Finance	2026-06-11
4	304	Sipho	HR	2026-06-11
5	305	Aisha	Marketing	2026-06-11

5 rows | 1.16s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 7:

Databricks SQL Editor interface showing a query execution result for 'online_orders'.

Query (Q7):

```
--Q7: online_orders
SELECT
  order_id,
  customer_id,
  order_date_text,
  TO_DATE(order_date_text, 'yyyy-MM-dd') AS order_date
FROM online_orders;
```

Table view showing 5 rows:

	order_id	customer_id	order_date_text	order_date
1	4001	101	2026-01-15	2026-01-15
2	4002	102	2026-02-20	2026-02-20
3	4003	103	2026-03-25	2026-03-25
4	4004	104	2026-04-10	2026-04-10
5	4005	105	2026-05-05	2026-05-05

5 rows | 0.88s runtime | See performance (1) | Checked for improvements | Refreshed now

QUESTION 8:

Databricks SQL Editor interface showing a query execution result for 'payment_dates'.

Query (Q8):

```
--Q8: payment_dates
SELECT
  payment_id,
  customer_id,
  payment_date,
  DATE_FORMAT(payment_date, 'yyyy-MM-dd') AS formatted_payment_date
FROM payment_dates;
```

Table view showing 5 rows:

	payment_id	customer_id	payment_date	formatted_payment_date
1	P001	101	2026-01-05	2026-01-05
2	P002	102	2026-02-10	2026-02-10
3	P003	103	2026-03-15	2026-03-15
4	P004	104	2026-04-20	2026-04-20
5	P005	105	2026-05-25	2026-05-25

5 rows | 1.24s runtime | See performance (1) | Checked for improvements | Refreshed now

QUESTION 9:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Catalog Focus ON + X

Run selected (1000) Just now (1) workspace: default New SQL editor: ON

```
--Q9 : customer_purchases
SELECT
  customer_id,
  customer_name,
  last_purchase_date,
  DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase
FROM customer_purchases;
```

Add parameter

	customer_id	customer_name	last_purchase_date	days_since_last_purchase
1	501	John	2026-05-01	41
2	502	Mary	2026-05-10	32
3	503	Peter	2026-05-15	27
4	504	Sarah	2026-05-20	22
5	505	Thabo	2026-05-25	17

5 rows | 1.35s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 10:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Catalog Focus ON + X

Run selected (1000) Just now (1) workspace: default New SQL editor: ON

```
--Q10:shipping_orders
SELECT
  order_id,
  customer_id,
  order_date,
  DATE_ADD(order_date,7) AS expected_delivery_date
FROM shipping_orders;
```

Add parameter

	order_id	customer_id	order_date	expected_delivery_date
1	6001	101	2026-05-01	2026-05-08
2	6002	102	2026-05-03	2026-05-10
3	6003	103	2026-05-05	2026-05-12
4	6004	104	2026-05-07	2026-05-14
5	6005	105	2026-05-09	2026-05-16

5 rows | 1.26s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 11:

databricks SQL Editor

Search data, notebooks, recents, and more... CTRL + P

workspace

SQL Editor

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q11: bookings
SELECT
  booking_id,
  customer_id,
  booking_date,
  YEAR(booking_date) AS booking_year,
  MONTH(booking_date) AS booking_month,
  DAY(booking_date) AS booking_day
FROM bookings;
```

Add parameter

	booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
1	B001	101	2026-01-12	2026	1	12
2	B002		102 2026-02-18	2026	2	18
3	B003		103 2026-03-22	2026	3	22
4	B004		104 2026-04-09	2026	4	9
5	B005		105 2026-05-27	2026	5	27

5 rows | 1.10s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 12:

databricks SQL Editor

Search data, notebooks, recents, and more... CTRL + P

workspace

SQL Editor

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q12: yearly_orders
SELECT
  order_id,
  customer_id,
  order_date,
  YEAR(order_date) AS order_year,
  amount
FROM yearly_orders
WHERE YEAR(order_date) = 2026;
```

Add parameter

	order_id	customer_id	order_date	order_year	amount
1	7004	104	2026-02-05	2026	1500.00
2	7005	105	2026-05-25	2026	2000.00

2 rows | 1.25s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 13:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON

Run selected (1000) Just now (1) workspace, default New SQL editor: ON

```
--Q13:monthly orders
SELECT
  order_id,
  customer_id,
  order_date,
  MONTH(order_date) AS order_month,
  amount
FROM monthly_orders
WHERE MONTH(order_date) = 3
```

Add parameter

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	3	800.00
8004	104	2026-03-25	3	1500.00

2 rows | 0.89s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 14:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON

Run selected (1000) Just now (1) workspace, default New SQL editor: ON

```
--Q14: subscriptions
SELECT
  subscription_id,
  customer_id,
  start_date,
  LAST_DAY(start_date) AS month_end_date
FROM subscriptions
```

Add parameter

subscription_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

5 rows | 1.03s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 15:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON + X

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q15: campaign_sends
SELECT
  send_id,
  customer_id,
  send_date,
  DATE_TRUNC('MONTH', send_date) AS month_start_date
FROM campaign_sends;
```

Add parameter

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01T00:00:00.000+00:00
C002	102	2026-02-18	2026-02-01T00:00:00.000+00:00
C003	103	2026-03-22	2026-03-01T00:00:00.000+00:00
C004	104	2026-04-09	2026-04-01T00:00:00.000+00:00
C005	105	2026-05-27	2026-05-01T00:00:00.000+00:00

5 rows | 1.30s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 16:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON + X

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

```
--Q16: invoice_dates
SELECT
  invoice_id,
  customer_id,
  invoice_date,
  DATE_FORMAT(invoice_date, 'MMMM yyyy') AS invoice_month_year
FROM invoice_dates;
```

Add parameter

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

5 rows | 1.45s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 17:

The screenshot shows the Databricks SQL Editor interface. The query being executed is:

```
--Q17: customer_birthdays
SELECT
  customer_id,
  customer_name,
  date_of_birth,
  FLOOR( DATEDIFF(CURRENT_DATE(), date_of_birth) / 365) AS customer_age
FROM customer_birthdays;
```

The results table displays 5 rows of data:

	customer_id	customer_name	date_of_birth	customer_age
1	901	John	1998-05-10	28
2	902	Mary	1990-08-20	35
3	903	Peter	2002-03-15	24
4	904	Sarah	1985-12-01	40
5	905	Thabo	2000-07-30	25

The interface also shows a sidebar with navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. The bottom status bar indicates 5 rows, 0.35s runtime, and a checked status for improvements.

QUESTION 18:

The screenshot shows the Databricks SQL Editor interface. The query being executed is:

```
--Q18: weekend_orders
SELECT
  order_id,
  customer_id,
  order_date,
  DAYNAME(order_date) AS day_name,
  CASE
    WHEN DAYNAME(order_date) IN ('Saturday', 'Sunday') THEN 'Weekend'
    ELSE 'Weekday'
  END AS day_type
FROM weekend_orders;
```

The results table displays 5 rows of data:

	order_id	customer_id	order_date	day_name	day_type
1	9001	101	2026-05-01	Fri	Weekday
2	9002	102	2026-05-02	Sat	Weekday
3	9003	103	2026-05-03	Sun	Weekday
4	9004	104	2026-05-04	Mon	Weekday
5	9005	105	2026-05-05	Tue	Weekday

The interface also shows a sidebar with navigation options like Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, and SQL Warehouses. The bottom status bar indicates 5 rows, 1.17s runtime, and a checked status for improvements.

QUESTION 19:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON

Catalog Type to search

- My organization
- Shares received

```

440
441 --Q19: quarterly_transactions
442 SELECT
443     transaction_id,
444     customer_id,
445     transaction_date,
446     QUARTER(transaction_date) AS transaction_quarter,
447     amount
448 FROM quarterly_transactions;

```

Add parameter

Table

	transaction_id	customer_id	transaction_date	transaction_quarter	amount
1	Q001	101	2026-01-15	1	500.00
2	Q002	102	2026-03-20	1	1200.00
3	Q003	103	2026-04-10	2	800.00
4	Q004	104	2026-07-05	3	1500.00
5	Q005	105	2026-10-25	4	2000.00

5 rows | 1.04s runtime See performance (1) Checked for improvements Refreshed now

QUESTION 20:

databricks Search data, notebooks, recents, and more... CTRL + P workspace

SQL Editor Focus: ON

Catalog Type to search

- My organization
- Shares received

```

451
452 --Q20: recent_orders
453 SELECT
454     order_id,
455     customer_id,
456     order_date,
457     DATEDIFF(CURRENT_DATE(), order_date) AS days_since_order,
458     amount
459 FROM recent_orders
460 WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;

```

Add parameter

Table

	order_id	customer_id	order_date	days_since_order	amount
1	R001	101	2026-04-01	71	500.00
2	R002	102	2026-04-15	57	1200.00
3	R003	103	2026-05-01	41	800.00
4	R004	104	2026-05-10	32	1500.00

4 rows | 0.91s runtime See performance (1) Checked for improvements Refreshed now

BONUS QUESTION:

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover Beta

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Spaces

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

SQL Editor

Catalog

Type to search

My organization

Shares received

Search data, notebooks, recents, and more... CTRL + P

workspace

JOINS

Date Functions

UNION AND UNION ALL

NULL FUNCTIONS

DATE FUNCTIONS

Run selected (1000) Just now (1s) workspace default New SQL editor: ON

462

463

464

465

466

467

468

469

470

471

472

473

-- BONUS CHALLENGE: customer_recency

SELECT

customer_id,

customer_name,

last_purchase_date,

DATEDIFF(CURRENT_DATE(), last_purchase_date) AS days_since_last_purchase,

CASE

WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 30 THEN 'Active Customer'

WHEN DATEDIFF(CURRENT_DATE(), last_purchase_date) <= 90 THEN 'At Risk Customer'

ELSE

INACTIVE Customer'

END AS customer_status

FROM customer_recency;

Add parameter

Table

+

customer_id

customer_name

last_purchase_date

days_since_last_purchase

customer_status

1

1001

John

2026-05-25

17

Active Customer

2

1002

Mary

2026-05-10

32

At Risk Customer

3

1003

Peter

2026-04-01

71

At Risk Customer

4

1004

Sarah

2026-02-15

116

Inactive Customer

5

1005

Thabo

2025-12-20

173

Inactive Customer

5 rows

1.46s runtime

See performance (1)

Checked for improvements

Refreshed now

-----THE END-----